NCERT Board-Style Solutions: the skill in one file (v1)

This page holds everything in the `skill/` folder in a form you can read from top to bottom, check, and pass on. The same material is split across `SKILL.md`, `references/` and `templates/` so that Claude can load it as a skill. If you change a rule here, change it in `references/format-rules.md` as well.

What it produces. For one chapter of an NCERT maths book: a LaTeX file and a compiled PDF with a solution for every worked example and every exercise question, using the exact textbook wording.

Who the solution is written for. A student whose own answer did not match. They may have gone wrong in the reasoning (wrong property, wrong triangle, wrong hypotenuse) or in the calculation (a sign, a square, a dropped factor, a forgotten last step). Every step has to be checkable by that student on their own.

Worked on so far: Class 10 Ch 7 (Coordinate Geometry) and Ch 10 (Circles). Both are drafts for the teacher's verification.

1. The rules

#	Rule	In practice
1	**Name the formula every time.**	Write it in general form before substituting: "By the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$." Properties count too: "the tangent is perpendicular to the radius through the point of contact."
2	**Output is LaTeX** plus the PDF.	Never only HTML or Word.
3	**Question, then one merged Solution that ends in a highlighted answer.**	`Question.` in bold, `Solution.` in bold, working, then common-mistake note, then verification, then the `\ans{...}` box. No separate "Answer." section. An alternative method, if any, is its own block *after* the box. Multi-part questions get one block, one box, per part.
4	**Show every value before it is squared, rooted or multiplied.**	One operation per line: formula with symbols, then the same line with numbers put in (negatives in brackets), then the differences, then the squares, then the sum, then the root. The same for algebra: show a division as a division, name an identity when you expand it, write $PR^2 = 4^2 = 16$ rather than jumping to 16.
5	**Common-mistake note** where a student is likely to slip.	`\slip{...}`: one or two sentences, what is done wrongly and why it is wrong. Leave it out where nothing plausible goes wrong.
6	**Exact NCERT wording**, including options (A)–(D).	Transcribed from rendered page images, never from the PDF text layer.
7	**Diagrams only when they help, never revealing, and matching the question.**	See section 3.
8	**Data read from a figure is an explicit setup step.**	"In Fig. 7.8 the horizontal axis is labelled Columns and the vertical axis Rows. Take columns along x and rows along y." Flag it to the teacher.

#	Rule	In practice
9	**Cover every question**: solved examples and all exercises.	Sub-parts (i), (ii) count separately.
10	**Main method, alternate method, verification** during the validation phase.	The verification shows its values.
11	**Proofs** use Given / To prove / Construction / Step–Statement–Reason table.	Each reason names the property in words. Ends in `\ans{...(Proved)}`.
12	**Units and ending.**	The `\ans{}` box carries the value with its unit, or the option letter and its value.
13	**Layout.**	No stranded headings, no `Overfull \hbox`, no wall-of-text opening paragraph (use a short list or a table).

2. What a finished solution looks like

This is the shape of Ex 10.1 Q3 in the Ch 10 book (figure omitted here):

```
\subsection*{Exercise 10.1, Question 3}
\textbf{Question.} A tangent $PQ$ at a point $P$ of a circle of radius $5$ cm meets a line
through the centre $O$ at a point $Q$ so that $OQ=12$ cm. Length $PQ$ is:\\
(A) $12$ cm \qquad (B) $13$ cm \qquad (C) $8.5$ cm \qquad (D) $\sqrt{119}$ cm.

\textbf{Solution.} Join $OP$. The tangent is perpendicular to the radius through the point of
contact, so $\ang OPQ=90\deg$. Hence $\tri OPQ$ is right-angled at $P$; its hypotenuse is $OQ$.

By the Pythagoras theorem, $OQ^2=OP^2+PQ^2$:
\begin{align*}
PQ^2&=OQ^2-OP^2\\
&=12^2-5^2\\
&=144-25=119\\
PQ&=\sqrt{119}\text{ cm}.
\end{align*}
\slip{The right angle is at $P$, so the hypotenuse is $OQ=12$ and we \emph{subtract}.
Adding, $12^2+5^2=169$, gives $13$, option (B): the tempting wrong choice.}

\textbf{Checking the other options.} ...
\textbf{Verification.} $OP^2+PQ^2=25+119=144=12^2=OQ^2$.

\ans{Option (D): $PQ=\sqrt{119}$ cm.}
```

For the distance formula the `\dist` macro prints all four rows (substitution, brackets, squares, sum and root) from the four coordinates, and TeX does the arithmetic, so the displayed numbers cannot disagree:

```
\distnote
\begin{align*}
\dist{PQ}{3}{2}{-2}{-3}{5\sqrt2\approx 7.07}\\
\distn{PR}{3}{2}{2}{3}
\end{align*}
```

gives, for the first row, $PQ = \sqrt{(-2-3)^2+(-3-2)^2} = \sqrt{(-5)^2+(-5)^2} = \sqrt{25+25} = \sqrt{50} = 5\sqrt{2} \approx 7.07$.

3. Figures

- **Never show what the question asks for.** No result lengths, solved coordinates or angle values.

- **Pure formula or derivation questions get no figure.**
- **If the book has a figure, redraw it**, caption it "(redrawn)", and mark only the givens.
- **Match the figure to the question.** After drawing, read the question word by word and tick each given against the picture: every named point and line drawn, every given length on the right segment, and the shape the solution relies on visible. In every calculation or MCQ figure that uses a right triangle, draw the right-angle mark (`\draw pic[draw,angle radius=0.3cm] {right angle=A-B-C};`, angle anticlockwise from `BA` to `BC`). In a proof figure leave it off when the 90° is a proof step. (Review round 2 found a missing right-angle mark in Ex 10.1 Q3, and a figure in Ex 7.2 Q3 that did not represent the field described.)
- **Construction lines** used in a proof are dashed and labelled "construction".
- **A construction question is the exception**: the figure is the answer.
- **If an honest figure would itself show the result** (a parallelogram around a circle is already a rhombus), omit it and write a one-line "Note on the figure".

4. Types of question and how to present each

#	Type	Typical wording	Core of the write-up
1	Numerical	"Find", "Calculate"	Named formula, values put in, simplification, verification, `\ans` with unit
2	Proof	"Prove", "Show that"	Given / To prove / Construction / Step–Statement–Reason table, then `\ans{... (Proved)}`
3	General derivation	"Derive"	Setup, symbols defined, each algebra step on its own line, boxed result
4	MCQ with justification	Options (A)–(D)	Work out the value, say why each other option fails and which slip leads to it, `\ans` gives letter and value
5	Fill in the blank	"Fill in the blanks"	One block per blank, a one-line reason, `\ans` holds the word
6	Reading data from a figure	"In Fig. ..."	Setup step stating axes and scale first, then calculate
7	Construction	"Draw", "Construct"	Numbered steps, finished figure, reasoning, one-line `\ans`
8	Word problem	A story with numbers	"Let ... be ..." first, solve, reject roots that do not fit, `\ans` in words
9	Verification	"Is it true that..."	Compute both sides separately, compare, state the conclusion

Method choice: use the textbook's method as the main one; add an alternate method only when it is genuinely different (similarity versus Pythagoras), not a re-wording; add a verification by substitution or a second route.

5. Step by step (the workflow)

1. **Get the source** (chapter PDF, learning plan if any) into `<class>/<chapter>/source/`.
2. **Render the pages and transcribe every question** from the images (section 6). Number every item in scope.
3. **Classify each item** using the table in section 4.
4. **Solve in Python first.** `source/check_numbers.py` asserts every numeric answer. No number goes into the LaTeX that the script has not confirmed. (Ch 7's script also re-checks the hand-typed value in every `\dist` row.)
5. **Write the LaTeX** with `templates/preamble.tex` and `templates/chapter_skeleton.tex`: one `.tex` per exercise and one for the examples, joined by `\input` in `main.tex`.
6. **Draw figures in TikZ** and check each against its question (section 3).
7. **Compile three times** with `pdflatex -interaction=nonstopmode main.tex`; the log must have no `^!`, `Overfull`, `Rerun` or `Undefined`.
8. **Look at every page** (`pdftoppm -r 70 -png main.pdf pg`) for overlapping labels, stranded headings, messy paragraphs, blank gaps and figures that reveal or mislead.
9. **Report** the file locations, every interpretation made, what was left out, and anything not solved with confidence.

6. Pitfalls when extracting questions from the PDF

1. Transcribe from rendered page images, not the text layer (it garbles $\angle$, $\perp$, θ, $\parallel$, $\triangle$, fractions, blanks and primes).
2. Square roots and fractions come out as plain text ("119" for $\sqrt{119}$): check against the image.
3. Fill-in blanks vanish from the text: write them as `\underline{\hspace{2cm}}`.
4. Options (A)–(D) may be reordered or merged: copy the order from the image.
5. "See Fig. 10.11": redraw from the image, keep the book's labels, caption "(redrawn)".
6. Strip running heads ("Mathematics", the chapter title) that the text layer inserts mid-question.
7. "Why?" prompts inside solved examples are questions to the student: answer them.
8. Theorems, Activities and "Think and Discuss" boxes are not exercises: leave them out and say so.
9. Count the questions per exercise from the images and compare with the solution file.
10. On Windows write `.tex` and `.py` files with a file-writing tool, not shell heredocs (they halve backslashes); `sed -i` may silently do nothing.
11. PyMuPDF (`import pymupdf`) can render pages if `pdftoppm` is missing.

7. Files and tools

```
math_solutions/
  skill/
    SKILL.md                          the skill entry point (loaded by Claude)
    NCERT_Board_Solutions_Skill_READ_ME.md   this file (source)
    NCERT_Board_Solutions_Skill_READ_ME.pdf  this file as a PDF, for reading
    references/  format-rules.md, solution-types.md, extraction-pitfalls.md
    templates/   preamble.tex, chapter_skeleton.tex, check_numbers_template.py
    tools/       readme_to_pdf.py  (rebuilds the PDF from the .md)
  class10/
    ch07_coordinate_geometry/  source/check_numbers.py, latex/main.pdf
    ch10_circles/              source/check_numbers.py, latex/main.pdf
```

Needed: `pdflatex` (MiKTeX or TeX Live) with `amsmath`, `amssymb`, `graphicx`, `enumitem`, `tikz`, `xcolor`, `tabularx`, `array`, `needspace`, `longtable`; `pdftoppm` (Poppler) or PyMuPDF; Python 3.

To hand to another teacher or another Claude chat: share the whole `skill/` folder (or the read-me PDF alone, for reading; the `.md` is the source that Claude and the build script use). To install as a Claude skill copy the folder to `~/.claude/skills/ncert-board-solutions/` (all projects) or `<project>/.claude/skills/ncert-board-solutions/`.

8. Open points to confirm with the teacher

- **Ch 7:** columns are taken as x and rows as y in the figure-reading questions; in Ex 7.2 Q3 the field is read with $AD = 100$ m; in Ex 7.1 Q6(ii) the answer is "no quadrilateral"; the TikZ figures are redraws and not to scale.
- **Ch 10:** Ex 10.2 Q11 has no figure on purpose (an accurate one would already be a rhombus); the Ex 10.1 Q4 figure is the answer by design; the Ex 10.2 Q5 figure has no dashed OP; theorems and activities are excluded.